Microeconomics

Course Code: Hum 2101

Lecture 1:Introduction to Economics

Reference Books:

- Paul A. Samuelson and William D. Nordhaus: Economics,18th edition, McGraw-Hill
- Michael Parkin: Microeconomics, 5th edition, Addison-Wesley.
- N. Gregory Mankiw: Principles of Microeconomics, 4th edition, Cengage Learning.

Meaning of the word Economics

- The word economy comes from the Greek word ***'oikonomos'***.

- This Greek word has two parts –

  - **'Oikos'** meaning *Home*

  - **'Nomos'** meaning *Management*

Hence, Economics means *Home Management* or *One who manages a Household*.

What is Economics?

Economics is the study of how societies use scarce resources to produce valuable goods and services and distribute them among different individuals.

Two key ideas in Economics -

1. Goods are scarce
2. Society must use its resource efficiently

Why Study Economics?

- You'll Expand Your Vocabulary
- You'll Understand Your Own Spending Habits
- You'll Improve Your Career Prospects

- Behavior of the financial markets
- why do some people or countries have high incomes while others are poor
- Business cycles—the fluctuations in credit, unemployment, and inflation
- Impacts of globalization
- How government policies can be used

Some Basic Concept

- **Needs**: It is something you must have. Needs refers to an individual's basic requirement that must be fulfilled, in order to survive. For example – air, water, etc.
- **Wants**: It is something you wish to have. Wants are described as the goods and services, which an individual like to have, as a part of his caprices. For example – a video game or a movie ticket.
- **Goods**: A physical product we can hold and touch.
- **Services**: An action provided to us.
- **Resources**: Anything used to get or create goods and services.

Free Goods and Economic Goods

Economic Goods: Economic goods are goods that are scarce and have a monetary value attached to them.

One of the key characteristics of economic goods is that they require some form of sacrifice or effort to acquire. This sacrifice can be in the form of money, time, or effort.

Free Goods: A free good does not require scarce resources for its production and does not have a monetary value attached to it.

One of the key characteristics of free goods is that they are often consumed without any direct cost to the consumer.

Scarcity and Efficiency

Scarcity: Scarcity means that society has limited resources and therefore cannot produce all the goods and services people wish to have.

Efficiency: Efficiency means that society is getting the maximum benefits from its scarce resources.

Economic efficiency requires that an economy produce the highest combination of quantity and quality of goods and services given its technology and scarce resources. An economy is producing efficiently when no individual's economic welfare can be improved unless someone else is made worse off.

Two Main Branches of Economics:

Microeconomics: Microeconomics is the study of how individual households and firms make decisions and how they interact with each other in the market.

Macroeconomics: Macroeconomics is the study of the economy as a whole with respect to output, price, land, employment, and other aggregate economic variable.

Microeconomics

- Adam Smith is usually considered the father of the field of Microeconomics.
- 16th June, 1723 – 17th July, 1790
- Adam Smith is a Scottish economist
- ***An inquiry into the Nature and Causes of the Wealth of Nations***

Macroeconomics

- John Maynard Keynes is the founder of Macroeconomics.
- 5th June, 1883 – 21st April, 1946
- British Economist
- Macroeconomics did not even exist in its modern form until 1936 when John Maynard Keynes published his revolutionary ***General Theory of Employment, Interest, and Money***.
- Everyone could grasp the importance of Macroeconomics or recognize it as a branch of economics after the Great Depression in 1930. Since 1936, macroeconomics developed as a separate strand within economics.

Differences between Microeconomics & Macroeconomics

Microeconomics

- Individual Markets
- Studies the Effect on Prices of Goods and Services
- Concerned with Individual Labour Markets & Consumer Behaviour
- Major factors include, Demand, Supply, Factor Pricing, Product Pricing, Production Consumption, etc.

Macroeconomics

- The Economy as a Whole
- Studies Economic Growth, National Income, Inflation, GDP, etc.
- Concerned with National, Territorial, and World Economies
- Major factors include, National Income, Distribution, Monetary Policies, Employment, General Price Level, Productive Capacity of Economy, etc.

The Logic of Economics:

1. ***The post hoc Fallacy***: The post hoc fallacy occurs when we assume that, because one event occurred before another event, the first event caused the second event.
2. ***Failure to hold other constants*** *or* Cateris Paribus *or* all other things remain constant *or* nothing else changes.
3. ***The fallacy of composition***: When you assume that what is true for the part is also true for the whole, you are committing the fallacy of composition.

Positive Economics and Normative Economics

- Positive economics is concerned with - what is
- Normative economics is concerned with - what should be

Positive Economics: Positive economics talks about things that exist. They are facts that can be verifiable. You can prove it or disprove it.

Normative Economics: Normative economics is fiction. They aren't facts; rather, they are economists' opinions who tell us what they think. It can be true for some and false for some.

Positive Economics and Normative Economics

POSITIVE ECONOMICS	VS	NORMATIVE ECONOMICS
Describes the facts without making value judgments or opinions.		Describes the facts and makes value judgments.
It's responsible for understanding economic phenomena.		Formulates and states the desirable reality.
How things are.		How thing should be.
Examples		**Examples**
• The current unemployment rate is 20%. • Health spending is 25%.		• The level of unemployment must be 5% maximum. • Defense spending cannot be greater than 10%.

Three Fundamental Problems of Economics

1. **What** goods and services are to be produced and in **What** quantities?

2. **How** should these goods and services be produced?

3. **For whom** these goods and services be produced?

Factors of Production

Inputs are commodities or services that are used to produce goods and services. An economy uses its existing technology to combine inputs to produce outputs.

Outputs are the various useful goods or services that result from the production process and are either consumed or employed in further production.

Another term for inputs is factors of production. These can be classified into four broad categories: ***land, labor, and capital & Organization or* entrepreneurship.**

Factors of Production

Land—or, more generally, natural resources, represents the gift of nature to our societies.

Labor consists of the human time spent in production working in automobile factories, writing software, teaching school, or baking pizzas.

Capital resources form the durable goods of an economy, produced in order to produce yet other goods. Capital goods include machines, roads, computers, software, trucks, steel mills, automobiles, washing machines, and buildings.

Organization - An entrepreneur is a person who combines the different factors of production (land, labor, and capital), in the right proportion and initiates the process of production, and also bears the risk involved in it.

Microeconomics

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Lecture 2: Circular Flow of Economics, Production Possibility frontier

Today we will learn -

- Circular Flow of Economics
- How to read Graphs
- Production Possibility frontier
- Opportunity Cost

Circular Flow of Economics

- **Two types of decision makers—firms, and households**

Firms produce goods and services using inputs, such as labor, land, and capital (buildings and machines). These inputs are called the factors of production.

Households own the factors of production and consume all the goods and services that the firms produce.

Circular Flow of Economics

- **Households and firms interact in two types of markets.**

Markets for goods and services

- households are buyers
- firms are sellers
- Households buy the output of goods and services that firms produce

Markets for the factors of production

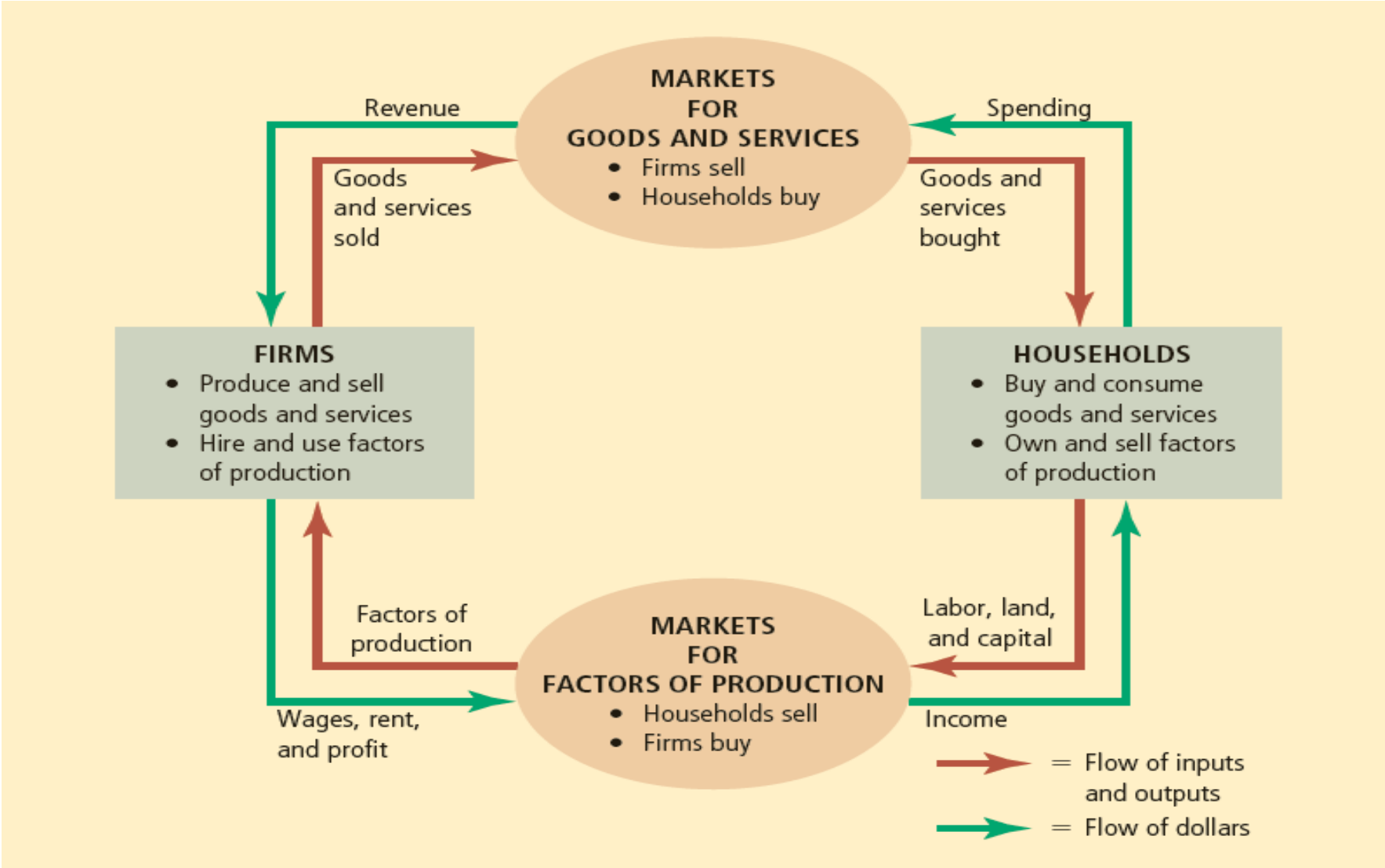
- households are sellers
- firms are buyers
- households provide the inputs that firms use to produce goods and services

Circular Flow of Economics

Two loops of the circular-flow diagram are distinct but related –

- The inner loop represents the flows of inputs and outputs.
- The outer loop of the diagram represents the corresponding flow of dollars.

Circular Flow of Economics



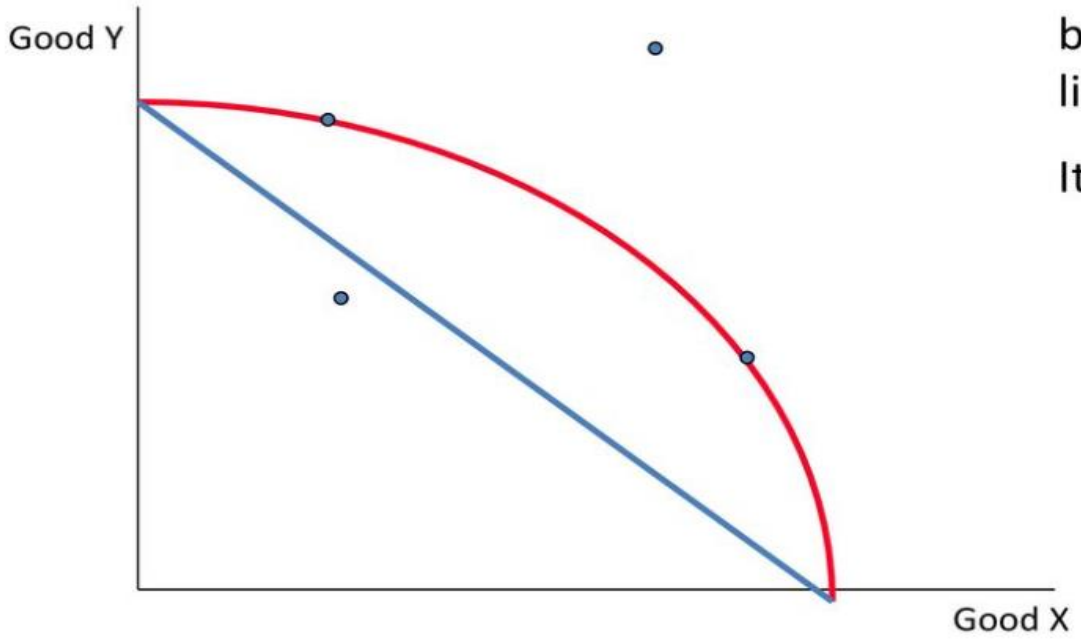
The Production Possibilities Frontier

The quantities of goods and services that we can produce are limited both by our available resources and by technology. And if we want to increase our production of one good, we must decrease our production of something else—we face a tradeoff.

The production possibilities frontier (PPF) is the boundary between those combinations of goods and services that can be produced and those that cannot.

The Production Possibilities Frontier

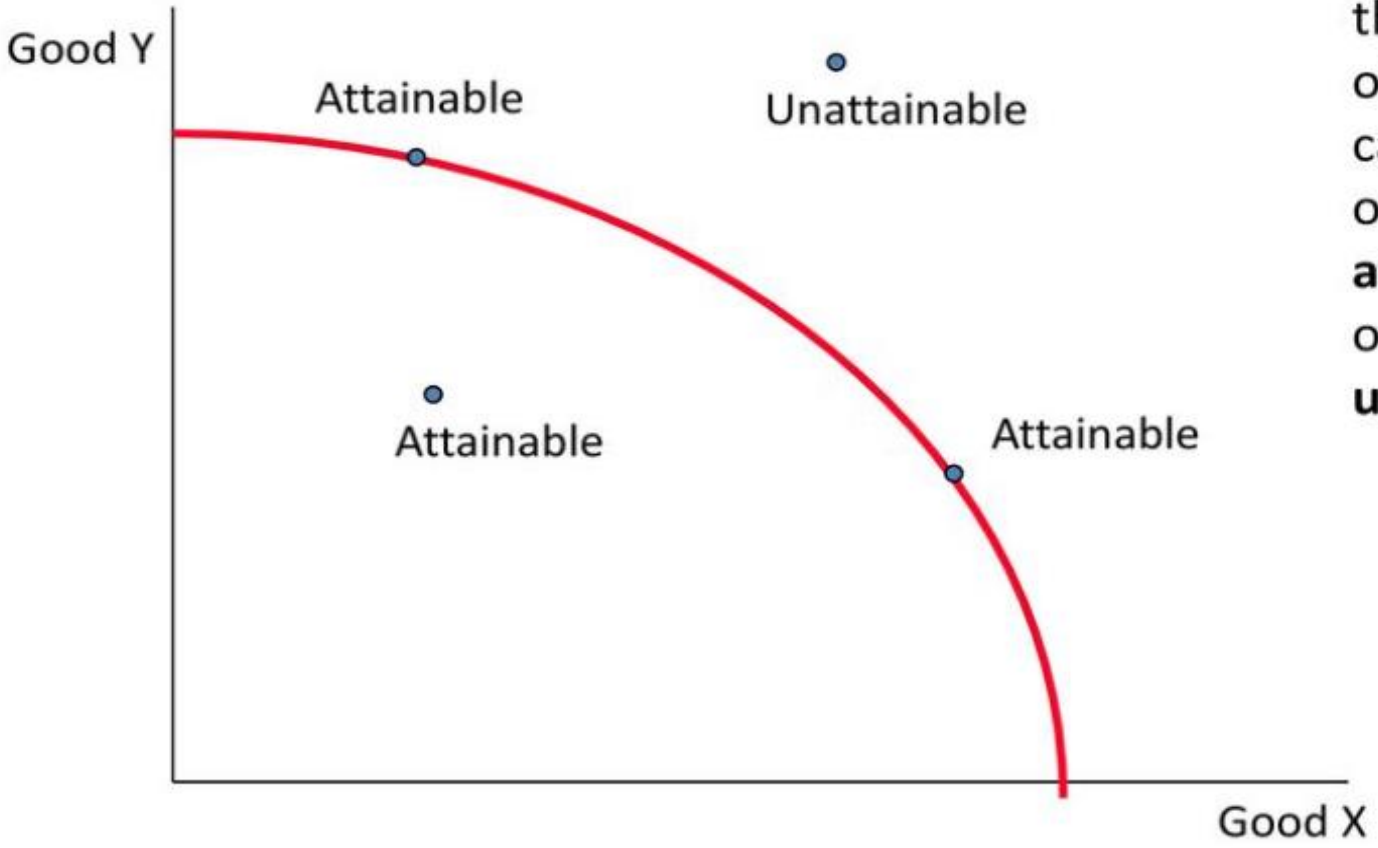
A Typical PPF Picture



The PPF is typically bowed-out or linear.

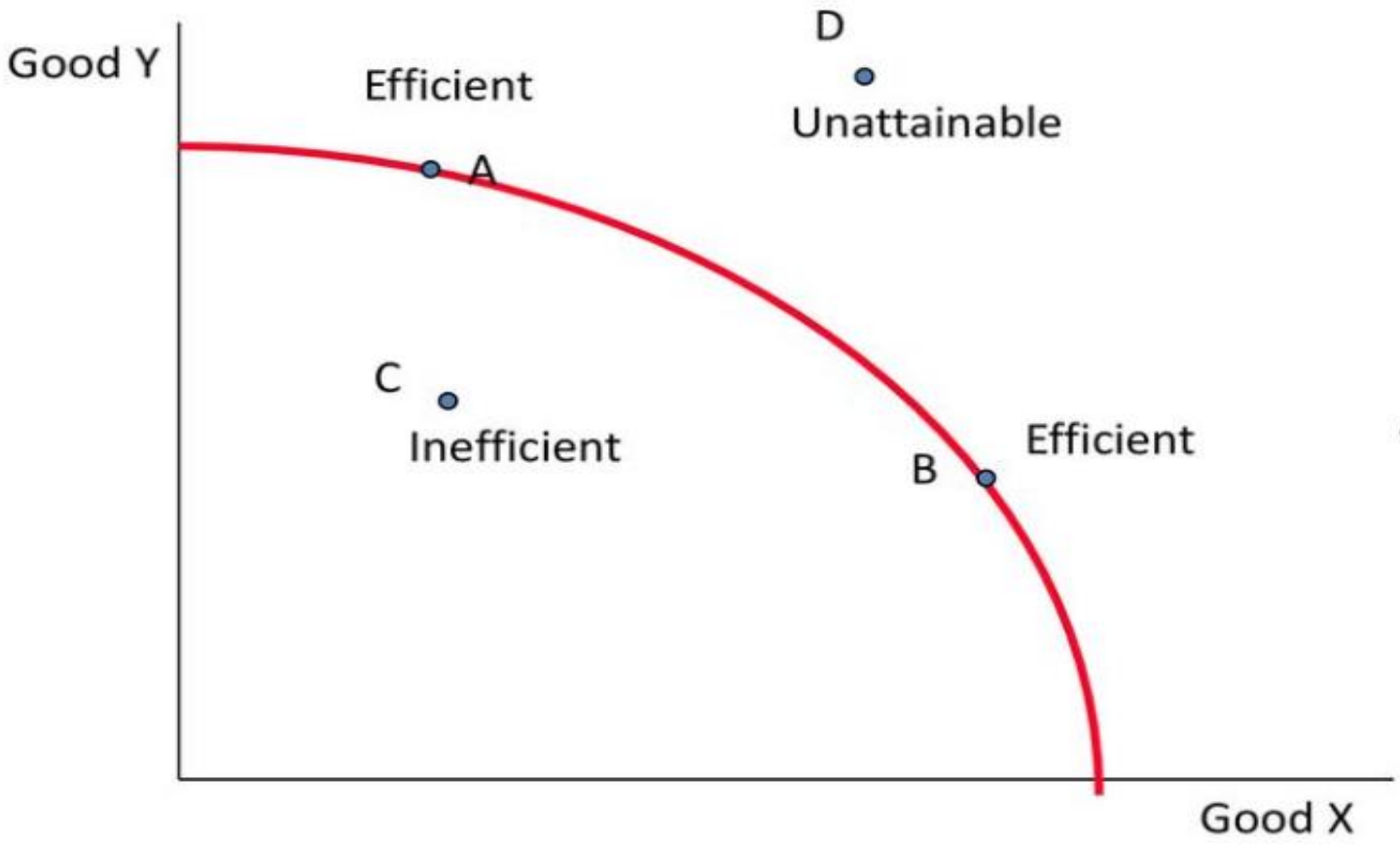
It is not bowed-in

The Production Possibilities Frontier



Since the PPF shows the limits to outputs of our economy, we can see directly, which outputs levels are **attainable** and which ones are **unattainable**.

The Production Possibilities Frontier



- According to the PPF points A and B - appearing on the curve – represent the most efficient use of resources by the economy.
- Point C represents an inefficient use of resources, while point D represents the goals that the economy can not attain with its present levels of resources.

The Production Possibilities Frontier

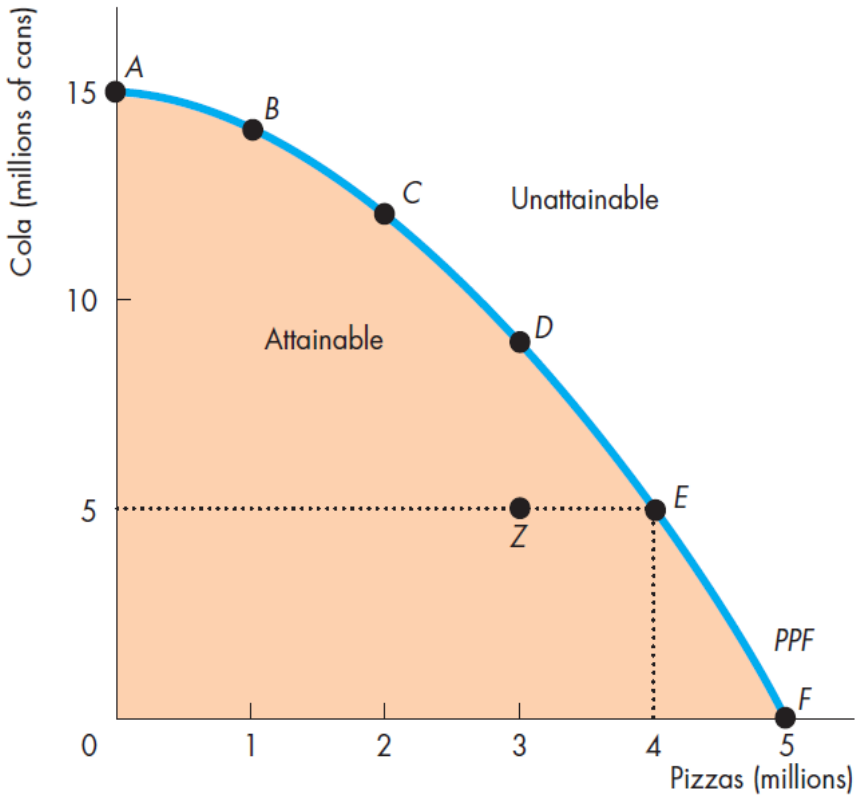
To illustrate the PPF,

- we focus on two goods at a time, and
- hold the quantities produced of all the other goods and services constant.

Let's look at the production possibilities frontier for cola and pizza, which represent any pair of goods or services.

The Production Possibilities Frontier

Possibility	Pizzas (millions)		Cola (millions of cans)
A	0	and	15
B	1	and	14
C	2	and	12
D	3	and	9
E	4	and	5
F	5	and	0

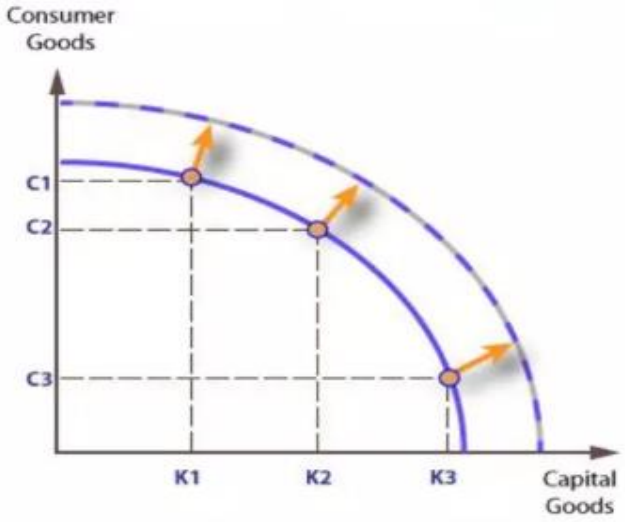


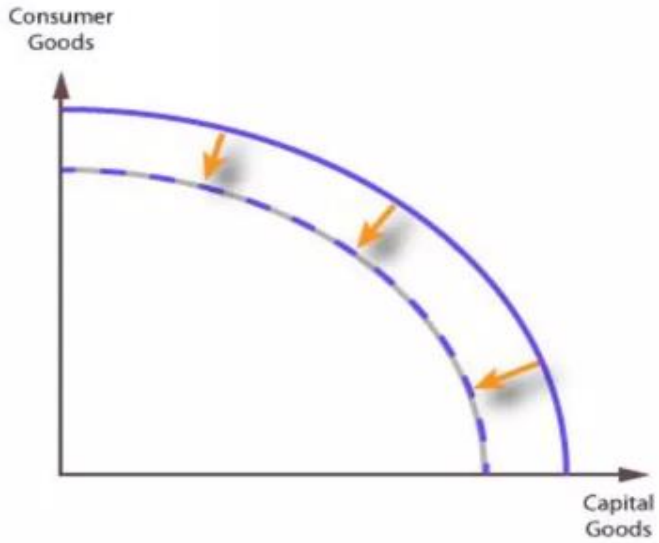
The Production Possibilities Frontier

Some observations from the graph:

- ➢The x-axis shows the quantity of pizzas produced
- ➢The y-axis shows the quantity of cola produced.
- The PPF separates the *attainable* from the unattainable.
- Production is possible at any point inside the orange area or on the frontier. Points outside the frontier are *unattainable*.
- Points inside the frontier, such as point Z, are *inefficient.*

The Production Possibilities Frontier





Outward Shift

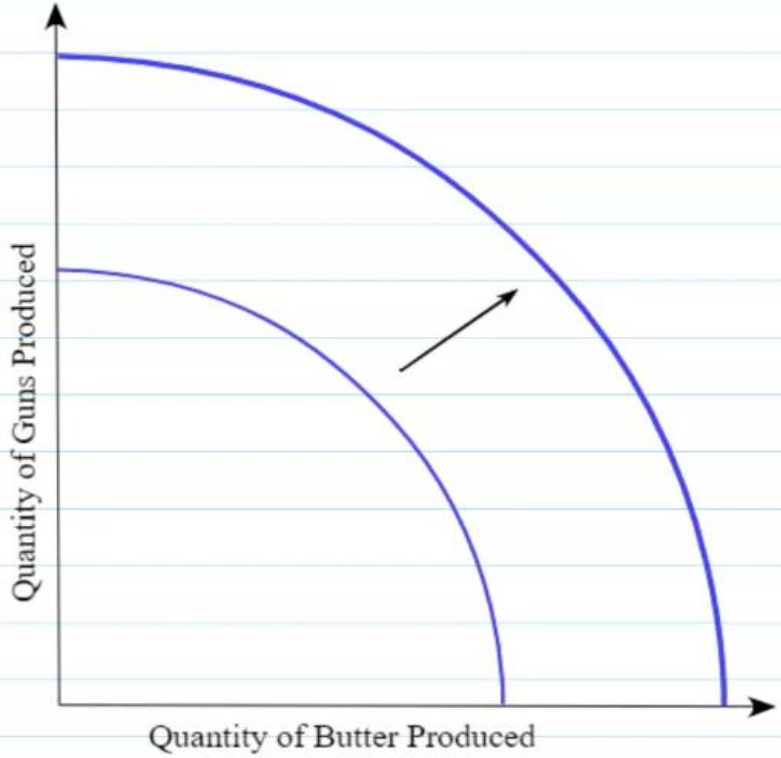
- New Technology
- New production method
- New Raw materials
- Increase in labor force

Inward Shift

- Resources run out
- Natural disaster
- Failure to invest
- Erosion of infrastructure

The Production Possibilities Frontier

What happens when technological advancement occurs?



- Shifts the PPFC rightward
- Less costly inputs are needed

Production Efficiency

We achieve **production efficiency** if we produce goods and services at the lowest possible cost.

Production is inefficient inside the PPF because resources are either ***unused*** or ***misallocated*** or ***both***.

Tradeoff Along the PPF

- Every choice along the PPF involves a tradeoff.
- This boundary is the real world's production possibilities frontier, and it defines the tradeoffs that we must make.
- On our real-world PPF, we can produce more of any one good or service only if we produce less of some other goods or services.
- All tradeoffs involve a cost—an opportunity cost.

Warm up question:

Tell me about a choice that you made yesterday. What was the opportunity cost of that choice?

Opportunity Cost

In a world of scarcity, choosing one thing means giving up something else. The opportunity cost of a decision is the value of the good or service forgone.

Opportunity Cost

When economists refer to the “opportunity cost” of a resource, they mean the value of the next-highest-valued alternative use of that resource.

If, for example, you spend time and money going to a movie, you cannot spend that time at home reading a book, and you can’t spend the money on something else. If your next-best alternative to seeing the movie is reading the book, then the opportunity cost of seeing the movie is the money spent plus the pleasure you forgo by not reading the book….

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Lecture 3: Basic Elements of Demand and Supply

What is Demand?

If you demand something, then you

- Want it,
- Can afford it
- Plan to buy it.

Quantity Demanded

Quantity demanded is the number of units of a good that individuals are willing and able to buy at a particular price during a time period.

For example, suppose that you buy one cup of coffee a day. The quantity of coffee that you demand can be expressed as 1 cup per day, 7 cups per week, or 365 cups per year.

The law of demand

The law of demand states,

Other things remain the same, the higher the price of a good, the smaller the quantity demanded; the lower the price, the greater the quantity demanded.

Law of Demand

When the price goes up...

...the quantity demanded goes down.

NOTE: *The relationship between price and quantity is inverse.*

When the price goes down...

...the quantity demanded goes up.

Demand Curve and Demand Schedule

This relationship between price and quantity bought is called the demand schedule, or the demand curve.

Let’s look at a simple example.

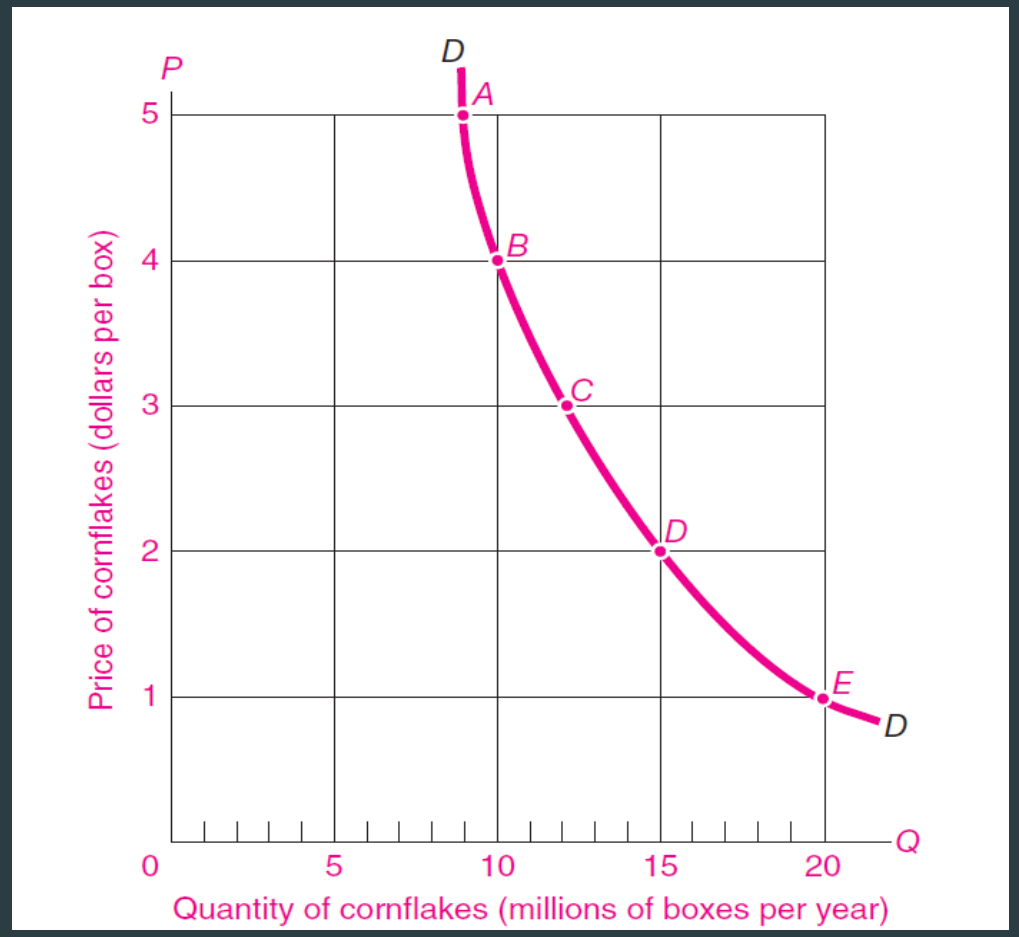
Demand Schedule

This table represents a hypothetical demand schedule for cornflakes-

Demand Schedule for Cornflakes

	(1) Price ($ per box) *P*	(2) Quantity demanded (millions of boxes per year) *Q*
A	5	9
B	4	10
C	3	12
D	2	15
E	1	20

Demand Curve



In the demand curve for cornflakes,

price (P) is measured on the vertical axis while quantity demanded (Q) is measured on the horizontal axis.

The negative slope of the demand curve illustrates the law of downward-sloping demand.

Law of downward-sloping demand

When the price of a commodity is raised (and other things are held constant), buyers tend to buy less of the commodity. Similarly, when the price is lowered, other things being constant, the quantity demanded increases.

Quantity demanded tends to fall as price rises for two reasons:

1. Substitution Effect
2. Income Effect

Law of downward-sloping demand

Substitution effect: It occurs because a good becomes relatively more expensive when its price rises.

For example, as the price of beef rises, I eat more chicken.

Income effect: This comes into play because when a price goes up, I find myself somewhat poorer than I was before.

If gasoline prices double, I have in effect less real income, so I will naturally curb my consumption of gasoline and other goods.

What is Supply?

If a firm supplies a good or service, the firm

- Has the resources and technology to produce it,
- Can profit from producing it
- Plans to produce it and sell it.

Quantity Supplied

The willingness and ability of sellers to produce and offer to sell different quantities of a good at different prices during a specific time period.

For example: When the price of an orange is 65 cents the quantity supplied is 300 oranges a week.

The Law of Supply

The law of supply states:

Other things remain the same, the higher the price of a good, the greater the quantity supplied; and the lower the price of a good, the smaller the quantity supplied.

Price
As Price Increases
Supply
Quantity supplied increases
Price
As price falls
Supply
Quantity supplied falls

Supply Curve and Supply Schedule

The supply schedule (or supply curve) for a commodity shows the ***relationship between its market price and the amount of that commodity that producers are willing to produce and sell, other things held constant***.

Let's look at a simple example.

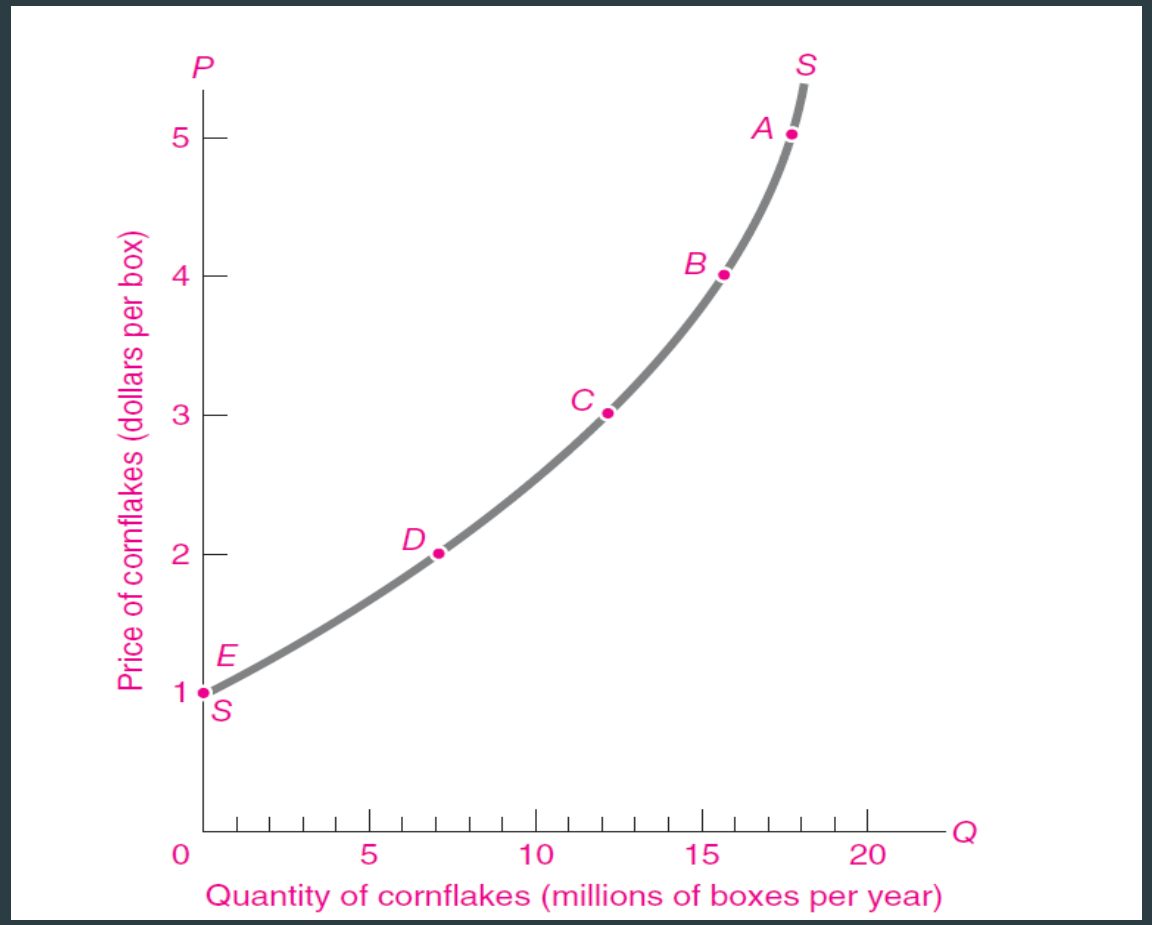
Supply Schedule

The table shows, for each price, the quantity of cornflakes that cereal makers want to produce and sell.

Supply Schedule for Cornflakes

	(1) Price ($ per box) *P*	(2) Quantity supplied (millions of boxes per year) *Q*
A	5	18
B	4	16
C	3	12
D	2	7
E	1	0

Supply Curve



The supply curve plots the price and quantity pairs from the supply schedule.

A smooth curve is passed through these points to give the upward-sloping supply curve, SS.

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Lecture 4: Changes in Demand & Supply

FACTORS AFFECTING THE DEMAND CURVE

- **Average income:** As incomes rise, people increase car purchases.
- **Population or Size of the market:** A growth in population increases car purchases.
- **Prices of related goods:** Lower gasoline prices raise car demand.
- **Tastes:** Having a new car becomes a status symbol.
- **Special influences:** Special influences include the availability of alternative forms of transportation, the safety of automobiles, expectations of future price increases, etc.

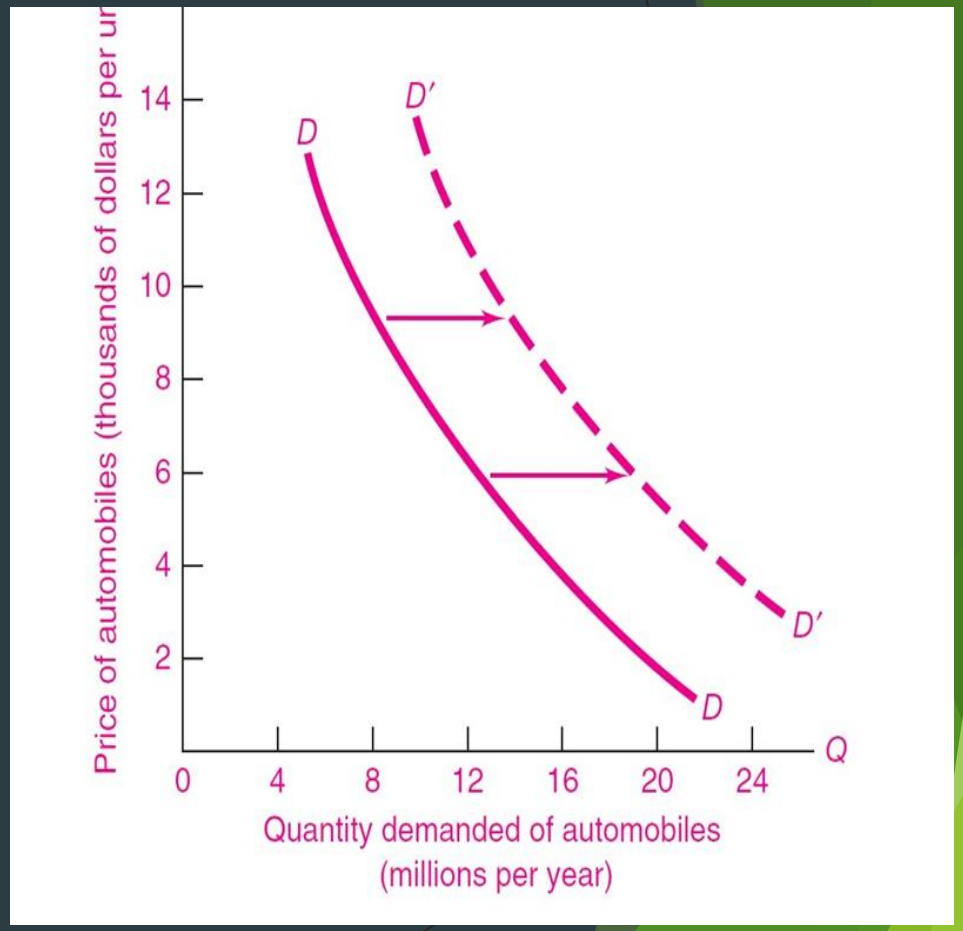
SHIFTS IN DEMAND

When there are changes in factors other than a good's own price which affect the quantity purchased, we call these changes shifts in demand. Demand increases (or decreases) when the quantity demanded at each price increases (or decreases).

SHIFTS IN DEMAND

Here we see the effect of rising average income, increased population, and lower gasoline prices on automobile demand. We call this shift of the demand curve an increase in demand.

Note that the shift means more cars will be bought at every price.

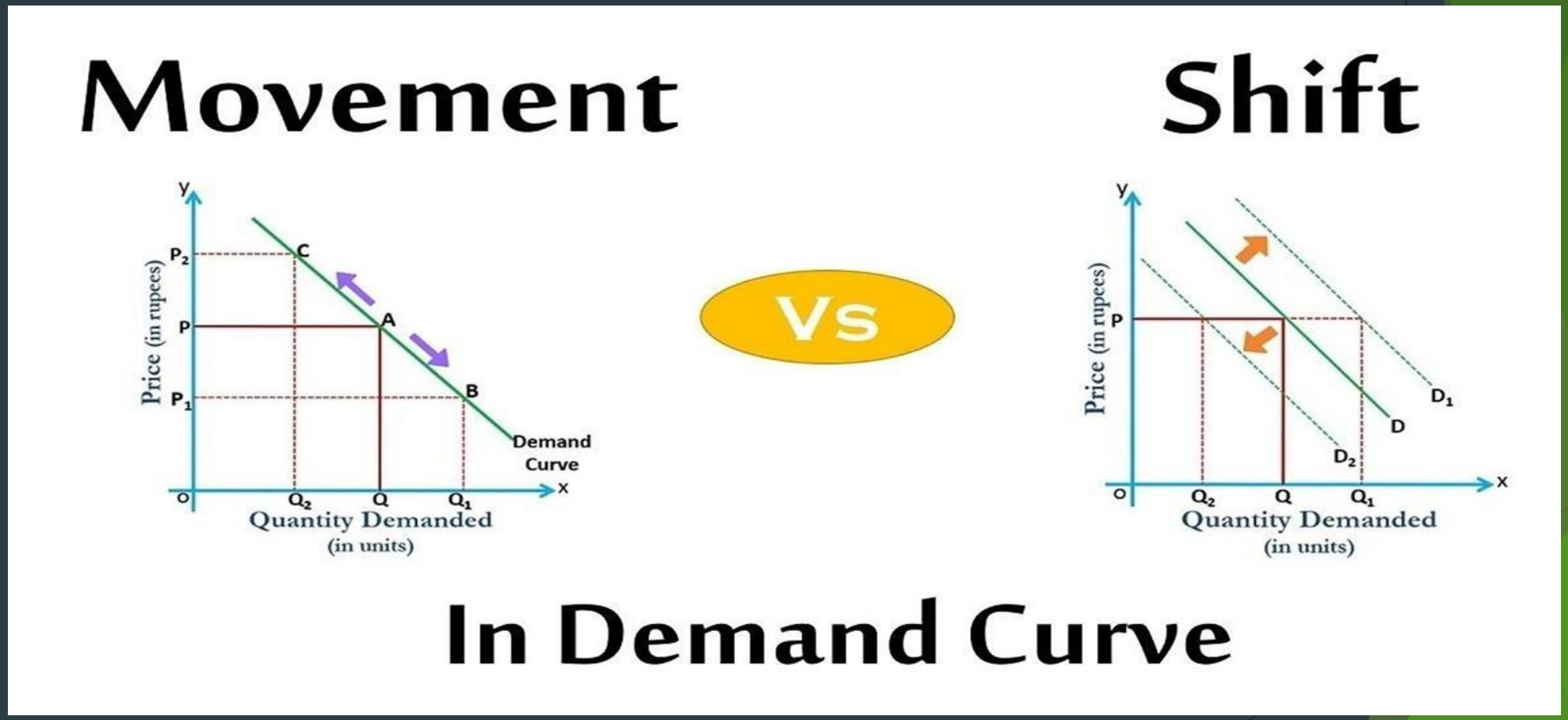


SHIFTS IN DEMAND

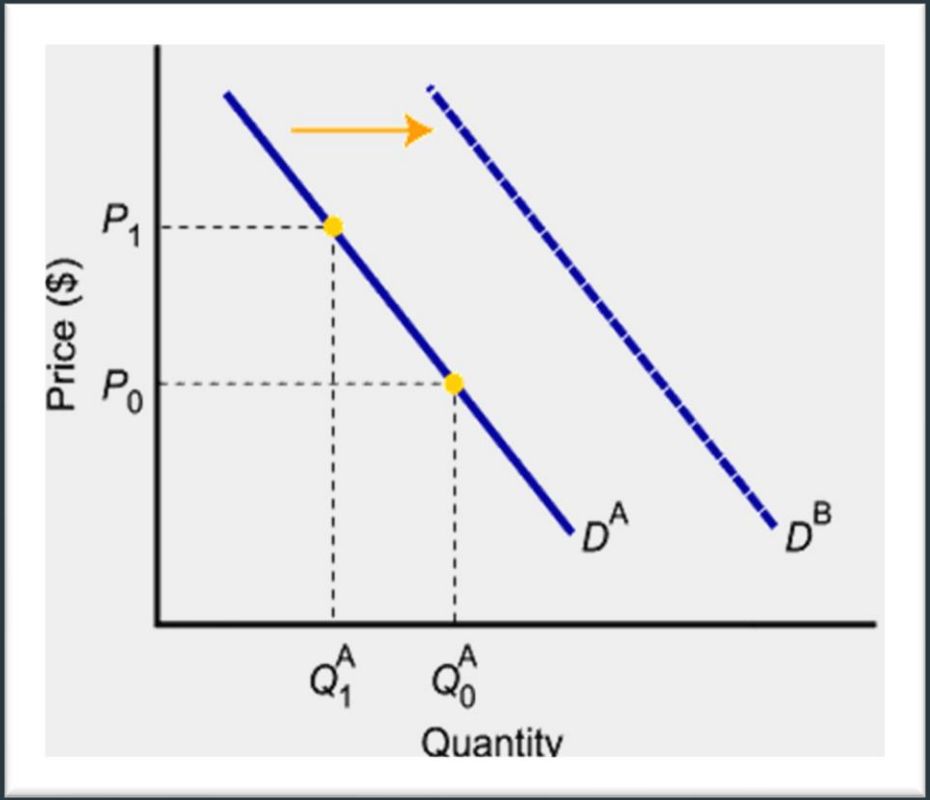
YOU CAN TEST YOURSELF BY ANSWERING THE FOLLOWING QUESTIONS:

- WILL A WARM WINTER SHIFT THE DEMAND CURVE FOR HEATING OIL LEFTWARD OR RIGHTWARD? WHY?
- WHAT WILL HAPPEN TO THE DEMAND FOR BASEBALL TICKETS IF YOUNG PEOPLE LOSE INTEREST IN BASEBALL AND WATCH BASKETBALL INSTEAD?
- WHAT WILL A SHARP FALL IN THE PRICE OF PERSONAL COMPUTERS DO TO THE DEMAND FOR TYPEWRITERS?
- WHAT HAPPENS TO THE DEMAND FOR A COLLEGE EDUCATION IF WAGES ARE FALLING FOR BLUE-COLLAR JOBS WHILE SALARIES FOR COLLEGE-EDUCATED WORKERS ARE RISING RAPIDLY?

MOVEMENTS ALONG CURVES VERSUS SHIFTS OF CURVES

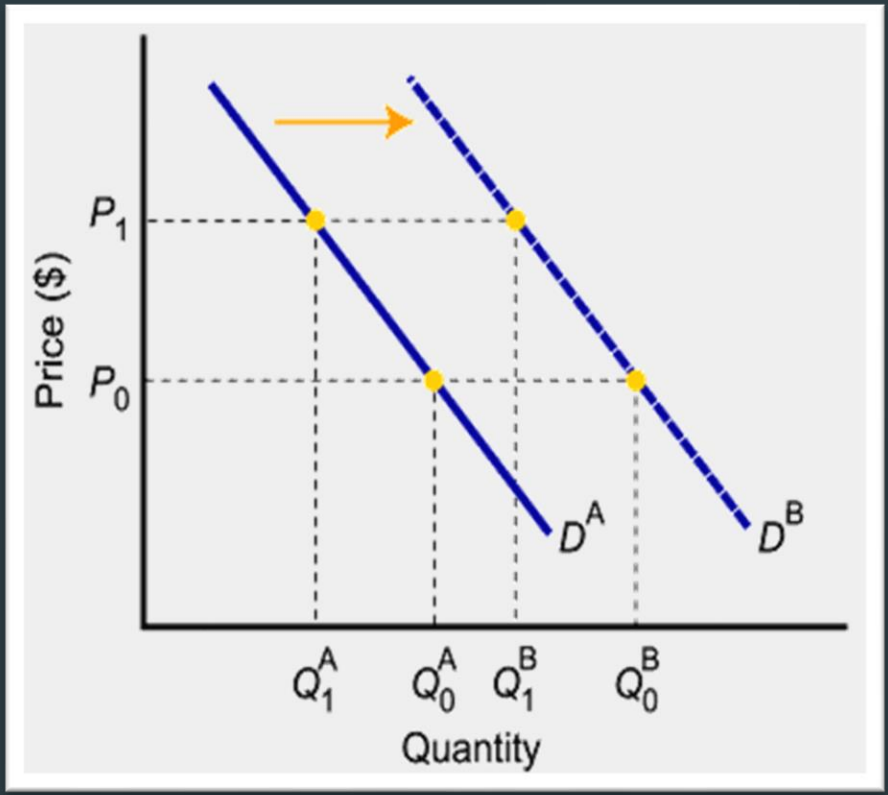


SHIFT OF DEMAND VERSUS MOVEMENT ALONG A DEMAND CURVE



- A change in demand is not the same as a change in quantity demanded.
- In this example, a higher price causes a lower quantity demanded.
- Changes in determinants of demand, other than price, cause a change in demand, or a shift of the entire demand curve, from DA to DB.

A CHANGE IN DEMAND VERSUS A CHANGE IN QUANTITY DEMANDED



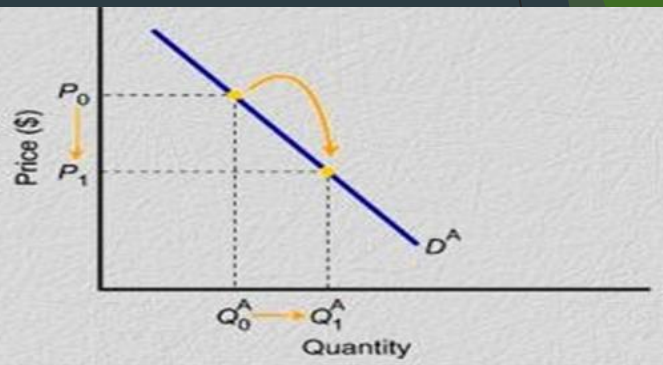
- When demand shifts to the right, demand increases. This causes the quantity demanded to be greater than it was before the shift, for every price level.

A CHANGE IN DEMAND VERSUS A CHANGE IN QUANTITY DEMANDED

TO SUMMARIZE:

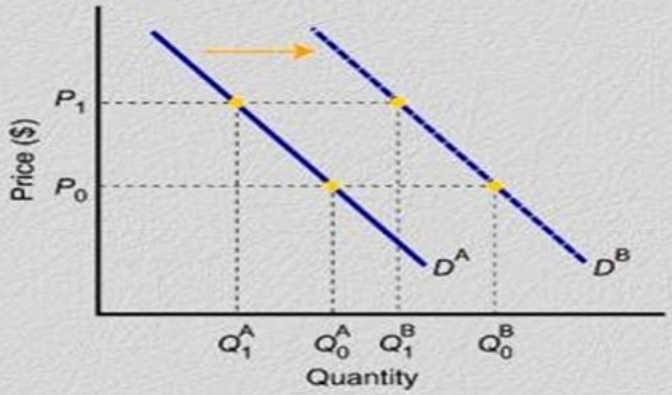
Change in price of a good or service leads to

Change in *quantity demanded* (**Movement along the curve**).



Change in income, preferences, or prices of other goods or services leads to

Change in demand (**Shift of curve**).



FACTORS AFFECTING THE SUPPLY CURVE

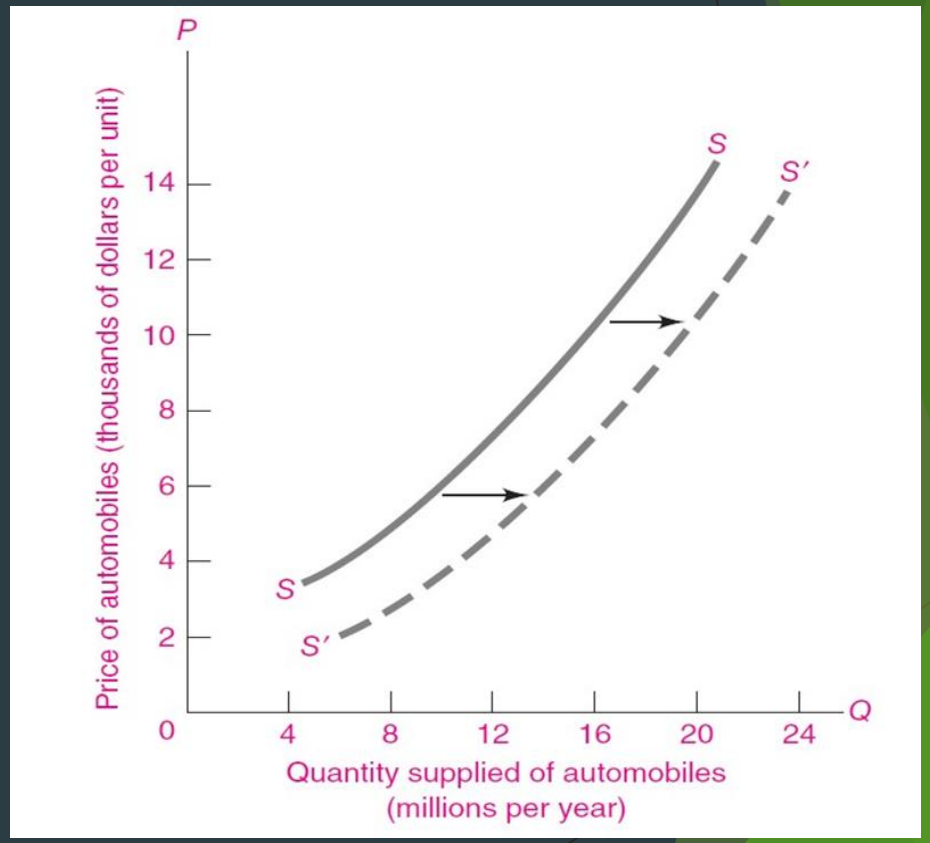
- Cost of production: When production costs for a good are low relative to the market price, it is profitable for producers to supply a great deal.
- Technology or Technological Advancement: Computerized manufacturing lowers production costs and increases supply.
- Input prices: A reduction in the wage paid to autoworkers lowers production costs and increases supply.
- Prices of related goods: If truck prices fall, the supply of cars rises.
- Government policy: Removing quotas and tariffs on imported automobiles increases the total automobile supply.
- Special influences: Internet shopping and auctions allow consumers to compare the prices of different dealers more easily and drive high-cost sellers out of business.

SHIFTS IN SUPPLY

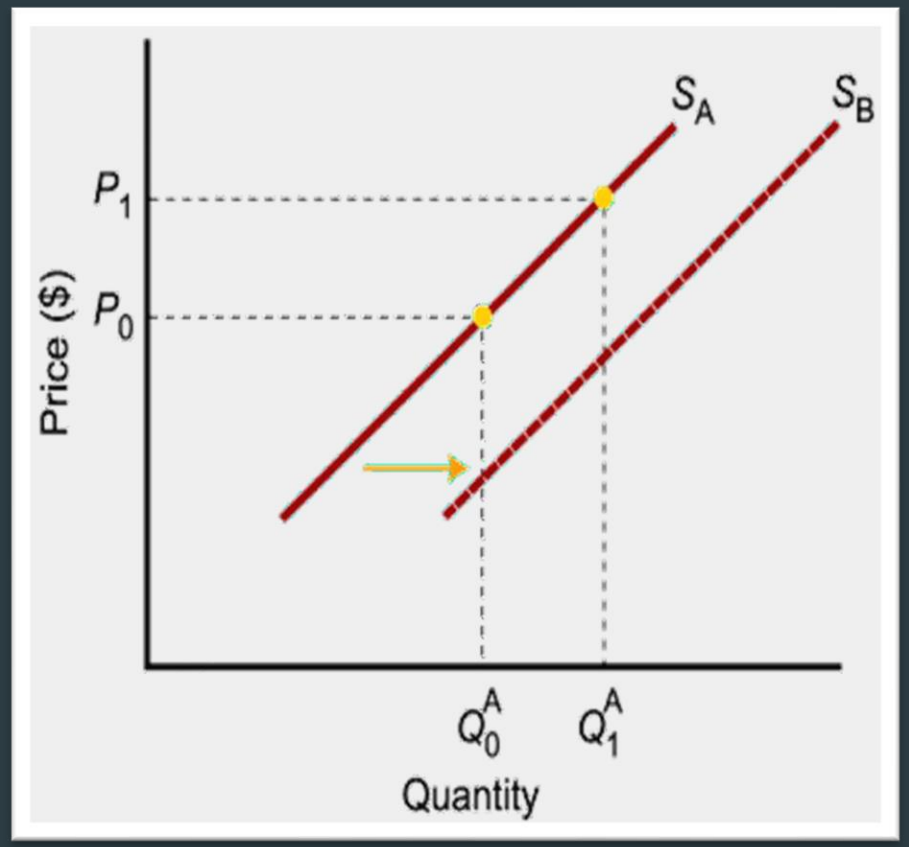
When changes in factors other than a good's own price affect the quantity supplied, we call these changes shifts in supply. Supply increases (or decreases) when the amount supplied increases (or decreases) at each market price.

SHIFTS IN SUPPLY

Here, as production costs fall, the supply of automobiles increases. At each price, producers will supply more automobiles, and the supply curve will shift to the right.

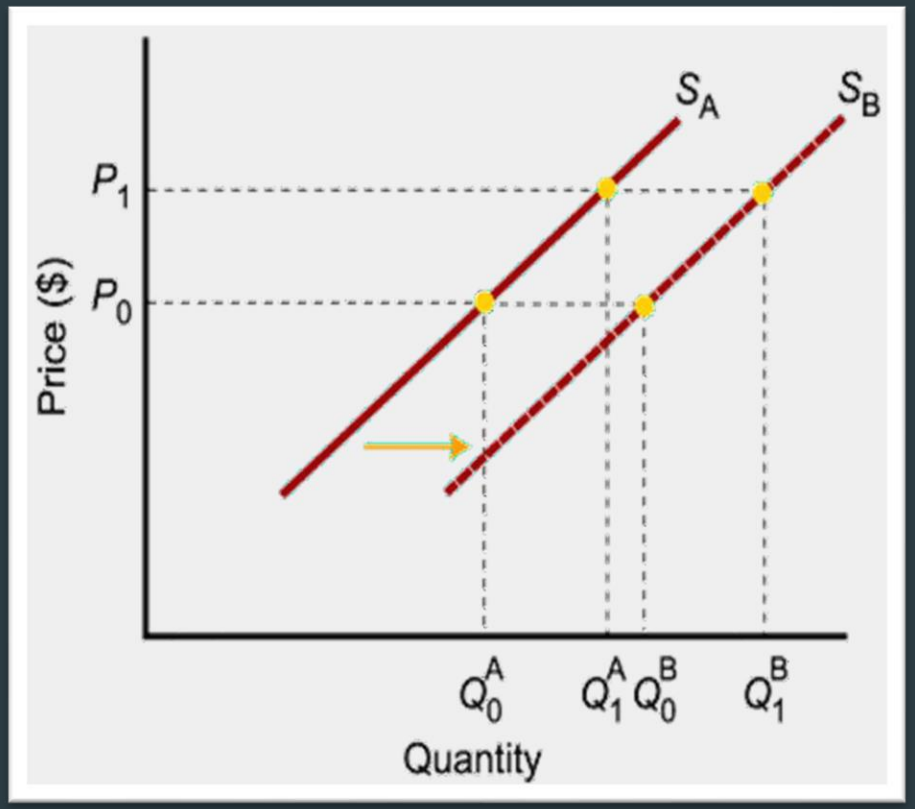


A CHANGE IN SUPPLY VERSUS A CHANGE IN QUANTITY SUPPLIED



- A change in supply is not the same as a change in quantity supplied.
- In this example, a higher price causes a higher quantity supplied, and a move along the supply curve.
- In this example, changes in determinants of supply, other than price, cause an increase in supply, or a shift of the entire supply curve, from SA to SB.

A CHANGE IN SUPPLY VERSUS A CHANGE IN QUANTITY SUPPLIED



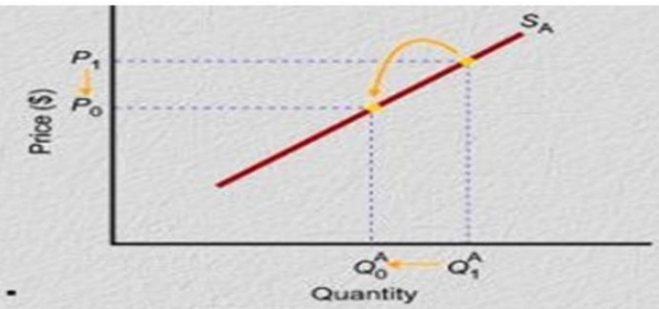
- When supply shifts to the right, supply increases. This causes the quantity supplied to be greater than it was prior to the shift, for each and every price level.

A CHANGE IN SUPPLY VERSUS A CHANGE IN QUANTITY SUPPLIED

TO SUMMARIZE:

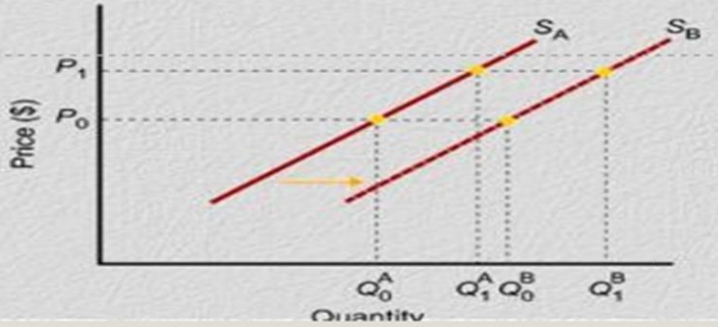
Change in price of a good or service leads to

Change in *quantity supplied* (**Movement along the curve**).



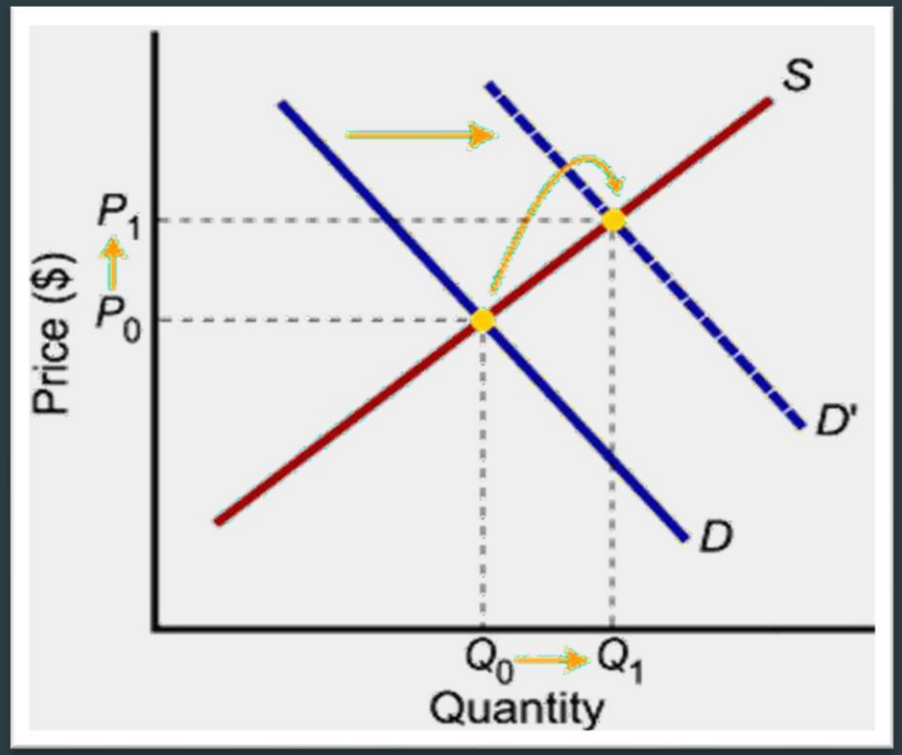
Change in costs, input prices, technology, or prices of related goods and services leads to

Change in supply (**Shift of curve**).

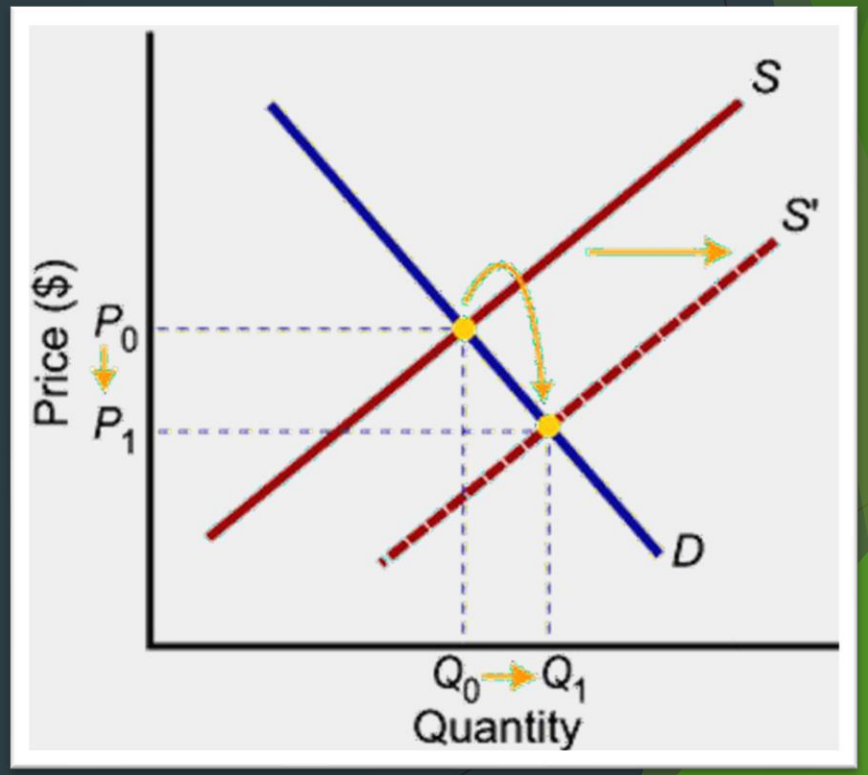


INCREASES IN DEMAND AND SUPPLY

HIGHER DEMAND LEADS TO HIGHER EQUILIBRIUM PRICES AND HIGHER EQUILIBRIUM QUANTITY.

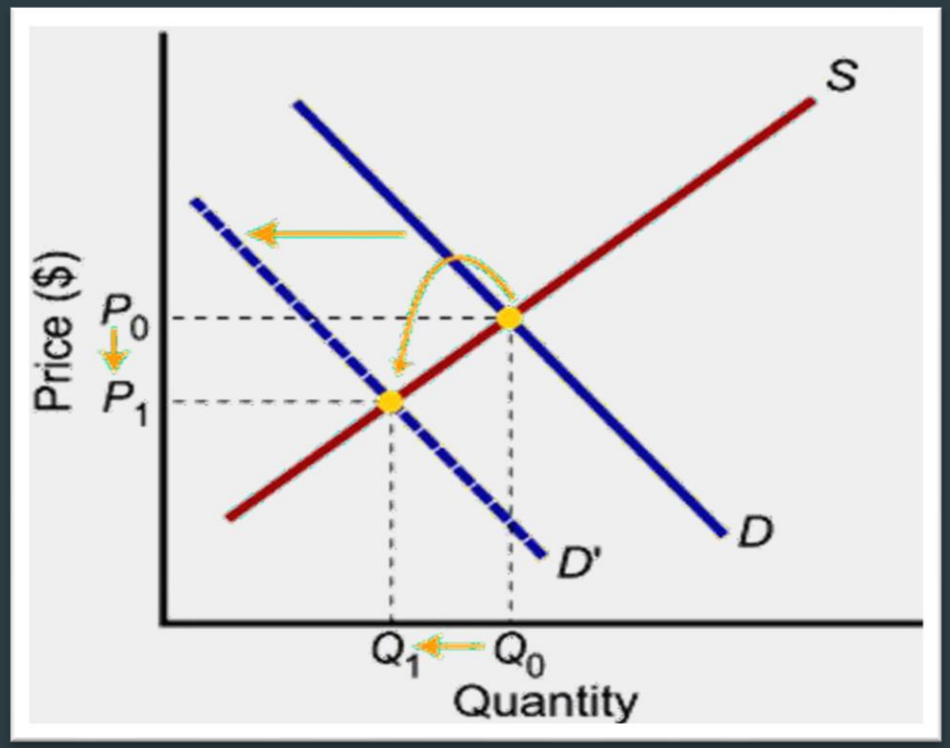
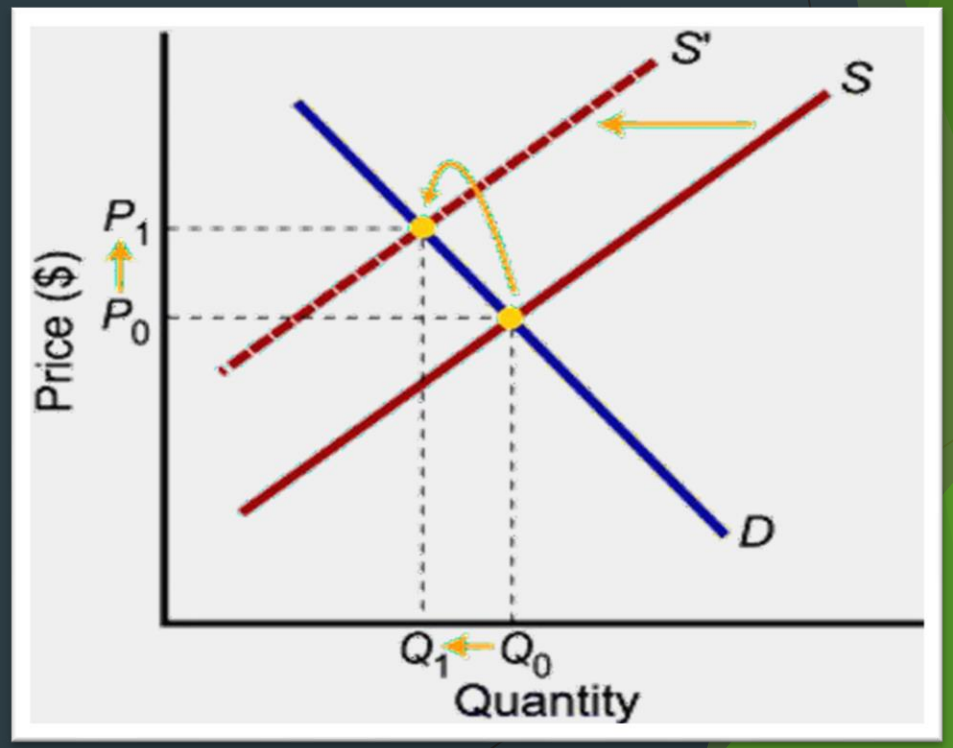


HIGHER SUPPLY LEADS TO LOWER EQUILIBRIUM PRICES AND HIGHER EQUILIBRIUM QUANTITIES.



DECREASES IN DEMAND AND SUPPLY

LOWER DEMAND LEADS TO LOWER PRICES AND LOWER QUANTITY EXCHANGED.



LOWER SUPPLY LEADS TO HIGHER PRICES AND LOWER QUANTITY EXCHANGED.